

Scheduler algorithm comparison

29 September 2026 | Decision brief and technical evidence

Decision summary

Retain the current booking selection and daily TABU default. Prioritize BOUNDED booking and CURRENT_UNCAPPED daily follow-up, investigate SHARED reliability, and evaluate KOPT by fleet size and available time. These modeled fixtures do not establish production savings.

BOUNDED confirmed 360/480 held-out requests versus INSERTION's 346/480, a 2.92 percentage-point gain. Its service advantage depends on concurrency. CURRENT_UNCAPPED has positive paired mean savings at every tested daily budget. KOPT's shorter-budget losses are concentrated in the largest fleet.

Coverage is the main limit: all 240 daily solver cases use CLUSTERED workloads. Only two held-out seeds (59 and 83) provide independent seed replication; they do not establish geographic diversity. Held-out booking uses dispersed GraphHopper 11 roads, but that coverage must not be attributed to daily solvers.

Experiment overview

Booking chooses where to admit a new request while preserving confirmed customer arrival windows. Daily optimization rearranges already accepted work without admitting demand. Internal planned arrival times and technician assignments may change while the customer promise remains valid. A completion flag describes search work, not whether a request was served or whether all possible schedules are infeasible.

Experiment	Retained scope	Comparison limit
Booking original	12 cases; earlier policy	Bundled policy/lifecycle changes; not isolated algorithm evidence
Booking screen	30 cases; clustered; 3 requests/case	Repeated geometry; one independent stream
Booking held out	144 cases; 2 seeds; 5/10/20/50 technicians; concurrency 1/5/10; cold/warm	48 cases and 480 requests per tested variant
Booking stress / browser	8 / 24 cases; 10 requests/case	Stress: INSERTION/BOUNDED; browser: INSERTION only
Daily	240 clustered cases: 48 screen + 192 held out	8 variants; held: 2 seeds x 4 fleets x 3 budgets

Mechanisms in plain language

Term	What the search does
Insertion	Places a new visit among existing stops without rearranging those stops.
Relocation / swap	Moves one visit to another position or route / exchanges two visits.
Reversal / sublist	Reverses an ordered stretch of visits / moves a stretch together.
Beam search	Retains a limited number of promising arrangements at each depth.
Tabu	Temporarily discourages revisiting recently changed entities to escape local traps.
Late acceptance	May accept a move if it beats an older solution, even when it worsens the current one.
K-opt	Reconnects several route links to explore coordinated order changes.
Ruin/recreate	Removes related visits, then rebuilds their placement to explore a larger change.

Modeled operating cost includes paid time and mileage. Retained rates are \$30/hour regular labor, \$45/hour overtime labor (new overtime prohibited), and \$0.67/mile. Travel buffers are 20% plus five minutes per leg. Waiting and buffers can change cost even with little or no road travel. These rates and fixture routes are assumptions, not measured payroll savings.

Booking configuration differences

Variant	Configuration and comparison
INSERTION	Policy-only control; eligible routes and insertion positions.
BOUNDED	Relocation/swap/reversal: 6 routes, depth 2, beam 8, 500 arrangements/window. Adds a search family, not one isolated move.
EXPANDED	12 routes, depth 3, beam 16, 2,000 arrangements/window and ranked moves. Multiple settings change together.
RUIN_RECREATE	EXPANDED plus complete reconstruction of removed related visits.
SHARED	Same reconstruction search with shared route evaluation across windows. The intended evaluation-sharing comparison is against RUIN_RECREATE in the screen.

Booking comparison



Figure 1. Separate categorical stages, with no connecting trend lines. Service counts confirmed requests; latency is the arithmetic mean of each case's p95, not an overall p95. Missing variants are untested. Original policy, stress, and browser populations differ.

Held variant	Served / rate	Unsuccessful	Incomplete	Served AND incomplete	Unknown completion
INSERTION	346/480 (72.1%)	134	132	0	0
BOUNDED	360/480 (75.0%)	120	120	0	0
SHARED	334/480 (69.6%)	146	153	7	0

Service and search completion overlap: 7 SHARED requests were served despite incomplete search. Unsuccessful means no confirmed service in this attempt; it does not prove infeasibility. Completion is reported separately, so incomplete counts must not be added to unsuccessful counts. SHARED's original final audit returned HTTP 409; its recovery run is separate.

Illustrative fixture evidence: rearrangement admits C

Technician		Before C	After C
omaha		A	C > B
nearby		B	A
Visit	Promised arrival window (UTC)		Planned arrival (UTC)
A	13:00 to 17:00 (end exclusive)		13:05
B	13:00 to 17:00 (end exclusive)		15:15
C	13:00 to 17:00 (end exclusive)		13:40

In reassignment-240-BOUNDED, A moves to the nearby technician and B to Omaha, allowing new request C to be served. Retained arrival plans remain inside the existing promises. The resulting plan has 100 driving minutes, 0 overtime minutes and \$183.31 modeled cost. The no-new-request oracle is \$91.66; the difference includes serving C and is not a savings estimate. This constructed fixture illustrates the mechanism, not its frequency in operations.

Held-out service by operating condition

Each breakdown pools the other tested dimensions within the same held-out population. It describes marginal outcomes, not an isolated causal effect. Percentage-point differences use INSERTION within the same group; all group denominators are equal across the three variants.

Concurrency

Concurrency	Variant	Served	Rate	Delta pp
1	INSERTION	160/160	100.00%	reference
1	BOUNDED	158/160	98.75%	-1.25
1	SHARED	160/160	100.00%	+0.00
5	INSERTION	96/160	60.00%	reference
5	BOUNDED	100/160	62.50%	+2.50
5	SHARED	96/160	60.00%	+0.00
10	INSERTION	90/160	56.25%	reference
10	BOUNDED	102/160	63.75%	+7.50
10	SHARED	78/160	48.75%	-7.50

Fleet size

Fleet size	Variant	Served	Rate	Delta pp
5	INSERTION	86/120	71.67%	reference
5	BOUNDED	91/120	75.83%	+4.17
5	SHARED	95/120	79.17%	+7.50
10	INSERTION	88/120	73.33%	reference
10	BOUNDED	92/120	76.67%	+3.33
10	SHARED	99/120	82.50%	+9.17
20	INSERTION	88/120	73.33%	reference
20	BOUNDED	89/120	74.17%	+0.83
20	SHARED	91/120	75.83%	+2.50
50	INSERTION	84/120	70.00%	reference
50	BOUNDED	88/120	73.33%	+3.33
50	SHARED	49/120	40.83%	-29.17

Cache condition

Cache condition	Variant	Served	Rate	Delta pp
cold	INSERTION	174/240	72.50%	reference
cold	BOUNDED	179/240	74.58%	+2.08
cold	SHARED	168/240	70.00%	-2.50
warm	INSERTION	172/240	71.67%	reference
warm	BOUNDED	181/240	75.42%	+3.75
warm	SHARED	166/240	69.17%	-2.50

At concurrency 1, BOUNDED serves 158/160 versus INSERTION's 160/160. BOUNDED serves more at concurrency 5 and 10. Its aggregate advantage does not mean it wins in every operating condition.

Latency and matched-customer cost

Request latency is the harness elapsedMs for the search, including queueing and polling, excluding selectionElapsedMs, address entry and geocoding. Request median and pooled p95 use all attempts, including unsuccessful ones. Pooled p95 uses nearest rank, $\text{ceil}(0.95 \times N)$. Mean case p95 retains the original statistic, giving each case equal weight; it is not the p95 of the pooled requests.

Stage	Variant	Requests	Mean case p95 s	Request median s	Pooled p95 s
original	INSERTION	18	0.61	0.50	0.85
original	BOUNDED	18	0.85	0.70	1.13
screen	INSERTION	18	0.75	0.58	0.97
screen	BOUNDED	18	3.37	2.82	4.08
screen	EXPANDED	18	20.03	18.80	25.95
screen	RUIN_RECREATE	18	23.37	19.01	31.44
screen	SHARED	18	19.51	17.49	25.40
held	INSERTION	480	2.88	1.02	4.57
held	BOUNDED	480	15.71	5.79	27.08
held	SHARED	480	64.15	32.77	121.03
stress	INSERTION	40	3.61	2.32	3.64
stress	BOUNDED	40	6.36	3.57	6.50
browser	INSERTION	240	3.53	1.63	5.43

Cost pairing requires identical served request indices on the same seed, fleet, workload, concurrency, cache condition and dataset fingerprint. First average matched case savings within a stream, then average streams equally. Clustered screen cases collapse to one stream; dispersed held-out streams are seed units. Do not extrapolate these subset estimates to all requests or production.

Stage	Vs INSERTION	Matched / tested cases	Stream units	Mean savings / stream
screen	BOUNDED	6/6	1	\$12.50
screen	EXPANDED	6/6	1	\$15.00
screen	RUIN_RECREATE	6/6	1	\$20.00
screen	SHARED	6/6	1	\$20.00
held	BOUNDED	15/48	2	\$23.44
held	EXPANDED	untested	untested	unavailable
held	RUIN_RECREATE	untested	untested	unavailable
held	SHARED	16/48	2	\$31.36

Search quality and booking implications

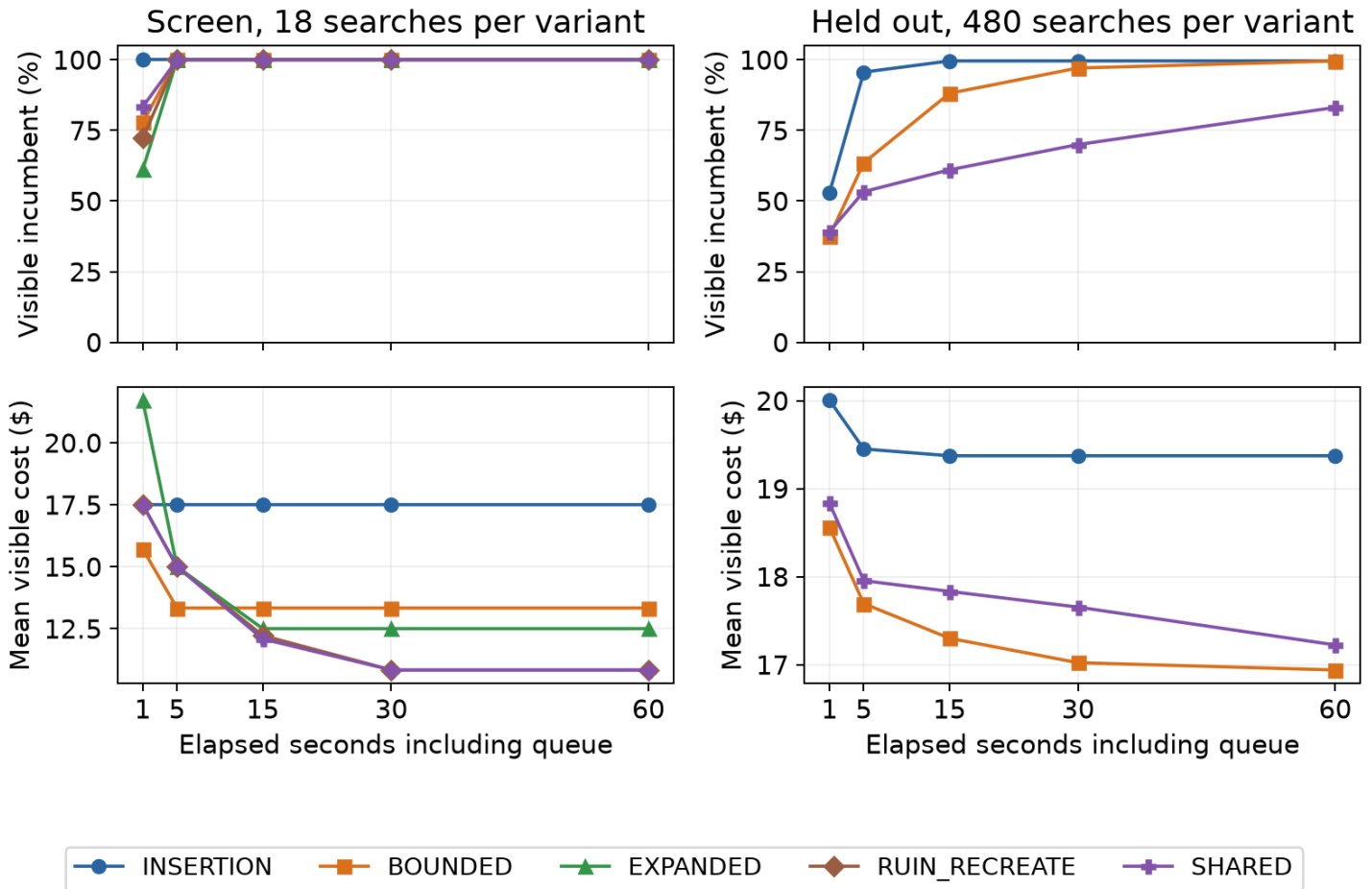


Figure 2. Fixed variant colors and markers across both stages, including SHARED in purple. Upper panels: fraction with a visible option by each checkpoint. Lower panels: incremental cost among visible options only; missing options are excluded, not zero. Visibility does not guarantee final confirmation.

Checkpoint time includes queueing and polling. Visible incremental cost is the option's modeled cost delta, not final total inventory cost. The lower panels have changing and variant-specific denominators, so they cannot rank all-request savings.

INSERTION: held service 346/480; mean case p95 2.9 s. Cost reference. Fast control and the only retained browser variant. Lower service than BOUNDED overall; retain as the current selection while testing changes.

BOUNDED: held service 360/480; mean case p95 15.7 s. Matched savings \$23.44/stream (15/48 cases). Best aggregate held-out service, but slower and not better at sequential service. Prioritize replicated concurrent and sequential tests together.

EXPANDED: screen service 18/18; mean case p95 20.0 s. Matched savings \$15.00/stream (6/6 cases). Screen-only broader beam and ranked moves. Higher matched cost savings than BOUNDED with substantially longer search; no held-out service claim is supported.

RUIN_RECREATE: screen service 18/18; mean case p95 23.4 s. Matched savings \$20.00/stream (6/6 cases). Screen-only reconstruction improves matched cost further. Its runtime and limited geography require a broader test before promotion.

SHARED: held service 334/480; mean case p95 64.1 s. Matched savings \$31.36/stream (16/48 cases). Matched cost savings coexist with lower service, long latency and a failed audit. Diagnose reliability before promotion.

Daily-solver comparison

All daily results report zero violations. Held-out comparisons pair the same dataset fingerprint, seed, fleet and budget, with identical starting costs. Accepted cost is measured after the policy's reference and fairness phases. Lower accepted cost means greater cleanup from that same start; different starting costs do not explain rankings here.

Configuration	What changes
CURRENT_CAPPED	Relocation/swap; 100 selected and accepted; 1,000-step cap.
CURRENT_UNCAPPED	Same moves/counts, without cap: isolates additional work.
LATE_ACCEPTANCE_CHANGE	Relocation only; history 400; 10,000 selected, 1 accepted. Changes acceptor and counts versus CURRENT.
LATE_ACCEPTANCE	Adds swap to CHANGE; same acceptor/counts.
TABU	Relocation/swap; entity tabu 7; 10,000 selected, 1,000 accepted. Changes multiple settings versus late acceptance.
SUBLIST	Late acceptance plus sublist movement/reversal.
KOPT	SUBLIST plus coordinated k-opt list moves.
RUIN_RECREATE	KOPT plus list ruin/recreate.

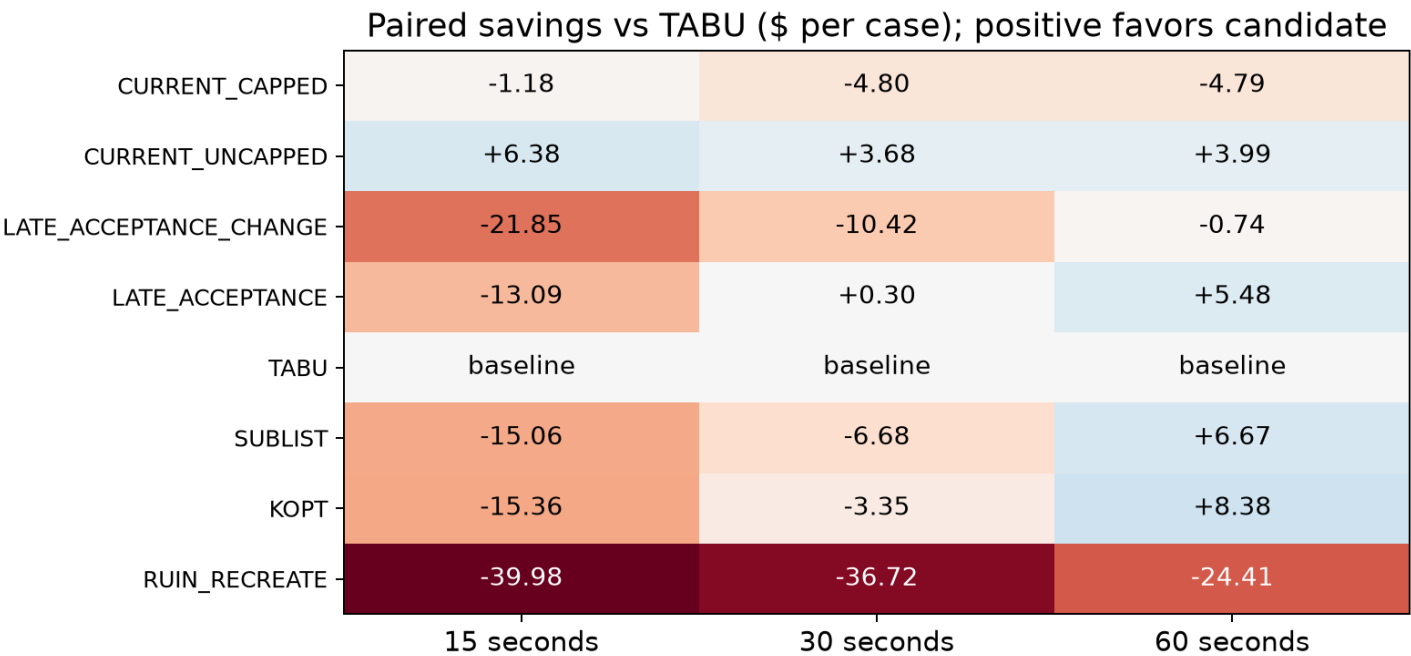


Figure 3. Held-out paired mean savings in dollars/case; 8 matched cases per cell (2 seeds x 4 fleet sizes). Blue is positive, red negative. Heatmap colors encode savings, not algorithm identity. Each seed averages its four fleets, then seeds receive equal weight.

Fleet size changes the daily conclusion

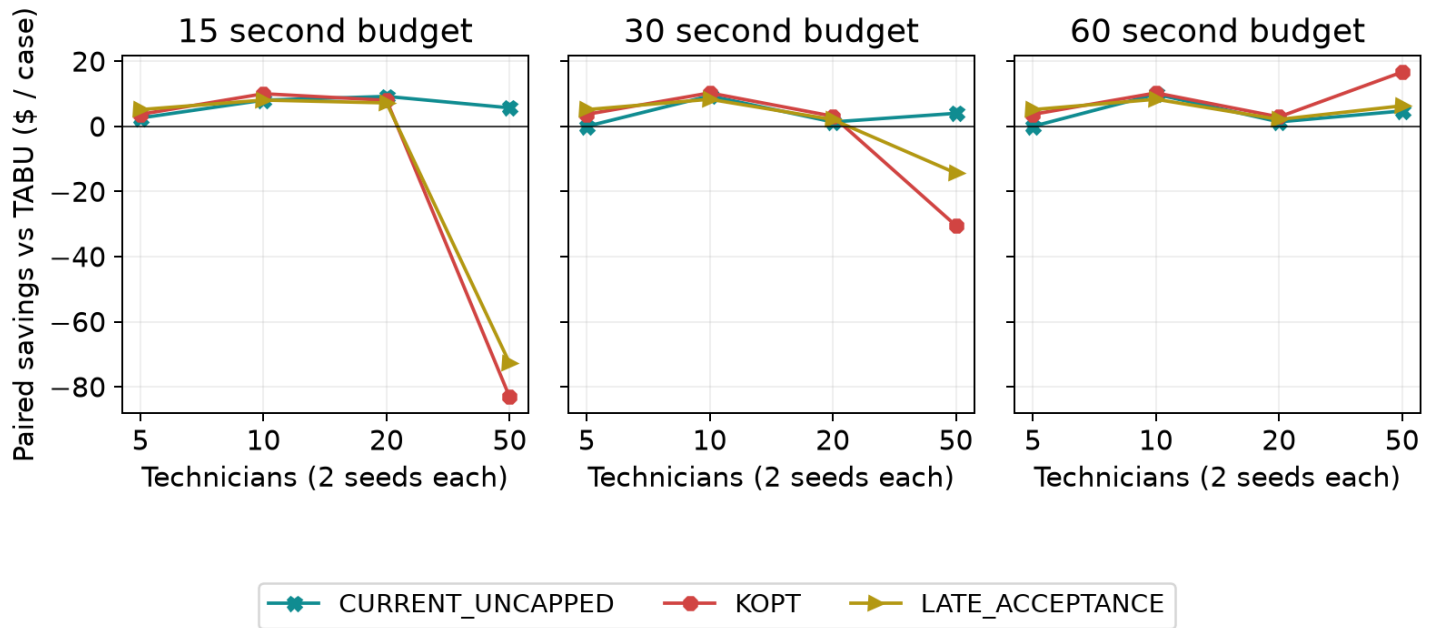


Figure 4. Selected follow-up candidates by fleet and budget; each point averages 2 matched seed cases. TABU is the zero reference. Equal spacing is categorical by fleet size. Full variant results follow in the appendix.

At 15 seconds, KOPT beats TABU in 6/6 smaller-fleet cases (5/10/20 technicians). It loses in 2/2 50-technician cases; their mean \$-83.16 paired savings drives the overall \$-15.36 mean.

At 30 seconds, KOPT beats TABU in 6/6 smaller-fleet cases (5/10/20 technicians). It loses in 2/2 50-technician cases; their mean \$-30.48 paired savings drives the overall \$-3.35 mean.

LATE_ACCEPTANCE already has a small positive mean at 30 seconds: \$0.30/case, rising to \$5.48 at 60 seconds. KOPT's largest overall advantage occurs at 60 seconds, but the shorter-budget smaller-fleet wins justify stratified follow-up.

Individual seeds and paired wins

Positive savings means the candidate's accepted cost is lower than TABU. Wins/ties/losses compare exact integer cents, with no tolerance. Each seed column averages four fleets, and W/T/L counts all eight paired cases per budget. Two seeds are insufficient for a broad population claim.

15-second budget

Variant	Seed 59 \$	Seed 83 \$	Mean \$	W/T/L (n=8)
CURRENT_CAPPED	-5.15	+2.80	-1.18	4/1/3
CURRENT_UNCAPPED	+2.35	+10.41	+6.38	6/1/1
LATE_ACCEPTANCE_CHANGE	-25.66	-18.05	-21.85	5/0/3
LATE_ACCEPTANCE	-16.25	-9.92	-13.09	6/0/2
TABU	0.00	0.00	0.00	0/8/0
SUBLIST	-16.82	-13.30	-15.06	6/0/2
KOPT	-17.73	-12.98	-15.36	6/0/2
RUIN_RECREATE	-42.82	-37.14	-39.98	6/0/2

30-second budget

Variant	Seed 59 \$	Seed 83 \$	Mean \$	W/T/L (n=8)
CURRENT_CAPPED	-8.00	-1.60	-4.80	3/1/4
CURRENT_UNCAPPED	-0.51	+7.88	+3.68	5/2/1
LATE_ACCEPTANCE_CHANGE	-10.60	-10.24	-10.42	5/0/3
LATE_ACCEPTANCE	-1.11	+1.71	+0.30	6/0/2
TABU	0.00	0.00	0.00	0/8/0
SUBLIST	-8.30	-5.07	-6.68	5/0/3
KOPT	-5.75	-0.94	-3.35	6/0/2
RUIN_RECREATE	-39.43	-34.02	-36.72	6/0/2

60-second budget

Variant	Seed 59 \$	Seed 83 \$	Mean \$	W/T/L (n=8)
CURRENT_CAPPED	-8.00	-1.59	-4.79	3/1/4
CURRENT_UNCAPPED	+0.05	+7.93	+3.99	5/2/1
LATE_ACCEPTANCE_CHANGE	-2.31	+0.83	-0.74	5/0/3
LATE_ACCEPTANCE	+3.26	+7.70	+5.48	8/0/0
TABU	0.00	0.00	0.00	0/8/0
SUBLIST	+5.13	+8.20	+6.67	7/0/1
KOPT	+5.59	+11.17	+8.38	8/0/0
RUIN_RECREATE	-25.02	-23.80	-24.41	6/0/2

Daily implications

All daily implications below are limited to two clustered held-out seeds. Each estimate has 8/8 matched cases per budget; mean gains can conceal seed and fleet differences.

CURRENT_CAPPED: paired savings at 15/30/60 seconds are \$-1.18, \$-4.80, \$-4.79 per case (8/8 matched at each budget). Speed reference: about three seconds regardless of allowance. Lower cost quality than TABU at all budgets; useful when little time is available.

CURRENT_UNCAPPED: paired savings at 15/30/60 seconds are \$6.38, \$3.68, \$3.99 per case (8/8 matched at each budget). Positive paired mean at every budget. Prioritize replication across dispersed and operational workloads; it consumes the full allowance.

LATE_ACCEPTANCE_CHANGE: paired savings at 15/30/60 seconds are \$-21.85, \$-10.42, \$-0.74 per case (8/8 matched at each budget). Relocation-only trajectory approaches TABU with more time but remains behind at each tested budget. No promotion signal.

LATE_ACCEPTANCE: paired savings at 15/30/60 seconds are \$-13.09, \$0.30, \$5.48 per case (8/8 matched at each budget). Adding swap improves on change-only at every budget; positive means at 30 and 60 seconds. Verify that advantage with broader workloads.

TABU: paired savings at 15/30/60 seconds are \$0.00, \$0.00, \$0.00 per case (8/8 matched at each budget). Retained production reference. Additional mean cleanup is flat from 30 to 60 seconds here; retain until broader evidence supports changing it.

SUBLIST: paired savings at 15/30/60 seconds are \$-15.06, \$-6.68, \$6.67 per case (8/8 matched at each budget). Positive at 60 seconds after shorter-budget losses. Longer-budget candidate; combined moves do not isolate an individual move effect.

KOPT: paired savings at 15/30/60 seconds are \$-15.36, \$-3.35, \$8.38 per case (8/8 matched at each budget). Largest overall 60-second mean advantage. Shorter-budget wins in smaller fleets are hidden by 50-technician losses; evaluate fleet and budget jointly.

RUIN_RECREATE: paired savings at 15/30/60 seconds are \$-39.98, \$-36.72, \$-24.41 per case (8/8 matched at each budget). Improves its own starting schedule but remains behind TABU. Larger neighborhood search does not repay its cost in this matrix.

Historical context: the separate September 24 comparison retained 294 accepted cases across seven workload families and three fleet sizes. All eight variants ran at seed 17; capped, uncapped and TABU repeated at seeds 23 and 41. TABU had lower aggregate accepted cost than both retained alternatives in every seed, supporting the default. Its configurations, policy and populations differ from September 29; do not pool the experiments. See the linked daily solver protocol for exclusions and original measurements.

Recommendations and follow-up

Priority / question	Additional experiment	Decision informed
1. BOUNDED service tradeoff	More independent dispersed streams, balanced across concurrency, fleet and cache; record confirmation failures and completion separately.	Whether its aggregate service gain survives the sequential shortfall and latency cost.
1. CURRENT_UNCAPPED generality	Matched TABU comparisons across dispersed, tight-window, skill and near-capacity work, retaining cost and fairness acceptance.	Whether to change the daily default.
2. SHARED reliability	Reproduce the retained HTTP 409 audit with complete client/server timing; trace served/incomplete overlap and final revalidation.	Whether the candidate is reliable enough for broader quality tests.
2. KOPT fleet/budget interaction	Replicated 5/10/20/50-technician comparisons at 15/30/60 seconds; preserve paired fingerprints.	Whether fleet-specific or longer-budget use is justified.
3. Longer-budget alternatives	Replicate LATE_ACCEPTANCE and SUBLIST beside KOPT and TABU on the same schedules.	Whether their cost/fairness/runtime tradeoffs merit promotion.
3. Expanded booking coverage	Test EXPANDED and RUIN_RECREATE on dispersed concurrent demand after reliability priorities.	Whether screen-only cost gains survive realistic service constraints.

No new benchmark runs, API changes or production configuration changes are part of this report revision. The recommendation remains to retain defaults pending the follow-up evidence.

Technical appendix: metric glossary

Metric	Definition and denominator
Operating cost	Modeled paid labor plus mileage under retained rates. Includes existing workload; final totals are not automatically savings.
Incremental cost	Visible option cost delta for adding a request; mean includes only searches with visible options at that checkpoint.
Paired booking savings	INSERTION final cost minus candidate final cost for identical served customers. Average within stream, then across streams; matched coverage shown beside estimate.
Paired daily savings	TABU accepted cost minus candidate accepted cost on the same case/budget. Average fleets within seed then seeds equally.
Served / unsuccessful	Confirmed request / attempt without confirmed service. Exhaustive and mutually exclusive for service only.
Incomplete / unknown	Explicit false / unavailable search-completion flag. Independent of service; never interpreted as proven infeasibility.
Visible option	At least one non-null cost incumbent by checkpoint. May fail final revalidation.
Mean case p95	Arithmetic mean of retained p95Ms for cases in a stage/variant. Not an overall request percentile.
Request median / pooled p95	Median / nearest-rank 95th percentile of all attempt elapsedMs within the stated group.
Fairness / waiting	Capacity-weighted utilization variance (lower is more balanced) / paid waiting minutes.
Schedule change	Changed technician assignments and retimed appointments, reported separately. Customer arrival promises remain constraints.
Untested / unavailable / zero	No experiment / metric cannot be estimated / measured numeric zero. These are not interchangeable.

Technical appendix: complete daily fleet comparison

Each cell is mean paired savings in dollars/case against TABU for exactly 2/2 matched seed cases at that fleet and budget. This appendix includes every variant, including the zero-cost reference.

15 seconds: savings by technician count

Variant	5	10	20	50
CURRENT_CAPPED	+2.57	+7.83	-2.60	-12.50
CURRENT_UNCAPPED	+2.57	+8.05	+9.20	+5.69
LATE_ACCEPTANCE_CHANGE	+4.69	+7.42	+1.82	-101.36
LATE_ACCEPTANCE	+5.14	+8.07	+7.20	-72.76
TABU	0.00	0.00	0.00	0.00
SUBLIST	+4.42	+9.55	+4.40	-78.61
KOPT	+3.69	+10.03	+8.00	-83.16
RUIN_RECREATE	+3.69	+10.25	+2.65	-176.51

30 seconds: savings by technician count

Variant	5	10	20	50
CURRENT_CAPPED	+2.57	+8.07	-10.25	-19.60
CURRENT_UNCAPPED	0.00	+9.38	+1.35	+4.00
LATE_ACCEPTANCE_CHANGE	+4.69	+7.67	-2.70	-51.34
LATE_ACCEPTANCE	+5.14	+8.32	+2.08	-14.36
TABU	0.00	0.00	0.00	0.00
SUBLIST	+4.42	+9.80	-1.23	-39.73
KOPT	+3.69	+10.28	+3.12	-30.48
RUIN_RECREATE	+3.69	+10.50	+0.38	-161.46

60 seconds: savings by technician count

Variant	5	10	20	50
CURRENT_CAPPED	+2.57	+8.07	-10.25	-19.57
CURRENT_UNCAPPED	0.00	+9.88	+1.38	+4.72
LATE_ACCEPTANCE_CHANGE	+4.69	+7.67	-1.48	-13.85
LATE_ACCEPTANCE	+5.14	+8.32	+2.12	+6.32
TABU	0.00	0.00	0.00	0.00
SUBLIST	+4.42	+9.80	-0.60	+13.04
KOPT	+3.69	+10.28	+2.90	+16.66
RUIN_RECREATE	+3.69	+10.50	+1.65	-113.47

Technical appendix: accepted schedule diagnostics

Held-out arithmetic means over eight cases per variant/budget. Cost change is accepted after minus before (negative means cleanup). Variance is accepted capacity-weighted utilization variance; waiting is paid minutes after acceptance. Assignment and timing changes count affected appointments. All 240 retained daily cases report zero violations.

15-second diagnostics

Variant	Cost change	Elapsed s	Variance	Waiting min	Assigned	Retimed
CURRENT_CAPPED	\$-54.46	3.24	0.008377	7.00	65.75	79.25
CURRENT_UNCAPPED	\$-62.02	15.01	0.012830	14.50	65.12	78.38
LATE_ACCEPTANCE_CHANGE	\$-33.78	15.00	0.007126	10.12	70.00	75.50
LATE_ACCEPTANCE	\$-42.55	15.00	0.010876	12.62	67.88	76.50
TABU	\$-55.64	15.00	0.017485	9.38	63.50	79.00
SUBLIST	\$-40.58	15.01	0.016880	13.00	70.88	78.12
KOPT	\$-40.28	15.01	0.011271	11.12	70.38	77.12
RUIN_RECREATE	\$-15.66	15.02	0.013895	14.75	69.88	77.38

30-second diagnostics

Variant	Cost change	Elapsed s	Variance	Waiting min	Assigned	Retimed
CURRENT_CAPPED	\$-54.46	2.92	0.008377	7.00	65.75	79.25
CURRENT_UNCAPPED	\$-62.94	30.00	0.011419	6.00	66.50	79.25
LATE_ACCEPTANCE_CHANGE	\$-48.84	30.00	0.015817	16.88	70.25	77.62
LATE_ACCEPTANCE	\$-59.56	30.00	0.017330	17.25	67.88	77.88
TABU	\$-59.26	30.00	0.014413	6.88	64.38	78.38
SUBLIST	\$-52.58	30.00	0.015058	10.50	68.62	78.25
KOPT	\$-55.92	30.00	0.012707	13.75	71.12	77.38
RUIN_RECREATE	\$-22.54	30.01	0.010853	11.38	71.50	78.38

60-second diagnostics

Variant	Cost change	Elapsed s	Variance	Waiting min	Assigned	Retimed
CURRENT_CAPPED	\$-54.46	2.82	0.008377	7.00	65.75	79.25
CURRENT_UNCAPPED	\$-63.25	60.01	0.009092	7.62	67.12	79.12
LATE_ACCEPTANCE_CHANGE	\$-58.52	60.00	0.017492	12.50	68.88	77.75
LATE_ACCEPTANCE	\$-64.73	60.01	0.011970	18.00	68.38	77.50
TABU	\$-59.26	60.00	0.014413	6.88	64.50	78.38
SUBLIST	\$-65.92	60.00	0.016121	7.75	70.12	78.88
KOPT	\$-67.64	60.00	0.012173	10.38	69.38	78.62
RUIN_RECREATE	\$-34.85	60.01	0.009020	8.62	71.12	76.75

Technical appendix: exploratory intervals

Percentile bootstrap intervals resample only two independent held-out seeds after averaging fleets (daily) or matched cases (booking). They are exploratory, not reliable population confidence intervals. Screen booking has one stream, so no interval is available. Individual seed values in the main report should receive greater weight than these intervals.

Daily variant	15 s: mean [interval] \$	30 s: mean [interval] \$	60 s: mean [interval] \$
CURRENT_CAPPED	-1.18 [-5.15, +2.80]	-4.80 [-8.00, -1.60]	-4.79 [-8.00, -1.59]
CURRENT_UNCAPPED	+6.38 [+2.35, +10.41]	+3.68 [-0.51, +7.88]	+3.99 [+0.05, +7.93]
LATE_ACCEPTANCE_CHANGE	-21.85 [-25.66, -18.05]	-10.42 [-10.60, -10.24]	-0.74 [-2.31, +0.83]
LATE_ACCEPTANCE	-13.09 [-16.25, -9.92]	+0.30 [-1.11, +1.71]	+5.48 [+3.26, +7.70]
TABU	baseline	baseline	baseline
SUBLIST	-15.06 [-16.82, -13.30]	-6.68 [-8.30, -5.07]	+6.67 [+5.13, +8.20]
KOPT	-15.36 [-17.73, -12.98]	-3.35 [-5.75, -0.94]	+8.38 [+5.59, +11.17]
RUIN_RECREATE	-39.98 [-42.82, -37.14]	-36.72 [-39.43, -34.02]	-24.41 [-25.02, -23.80]
Booking stage / variant	Matched / tested	Mean / interval per stream	
screen / BOUNDED	6/6	\$12.50 / unavailable: one stream	
screen / EXPANDED	6/6	\$15.00 / unavailable: one stream	
screen / RUIN_RECREATE	6/6	\$20.00 / unavailable: one stream	
screen / SHARED	6/6	\$20.00 / unavailable: one stream	
held / BOUNDED	15/48	\$23.44 / \$23.28 to \$23.59	
held / SHARED	16/48	\$31.36 / \$29.84 to \$32.87	

Technical appendix: evidence and reproducibility

The inventory retains 218 validated booking cases and 240 daily solver cases. Every inventoried raw file is checked by SHA-256; compressed JSONL content is also checked by uncompressed SHA-256. The original failed SHARED seed-83 audit is retained and excluded from successful cases, with its recovery separately labeled. Eight original stress-normal-only audits omitted overflow dates and remain excluded; full-date reruns supply the validated stress set. No inclusion rules have changed.

Failure detail: the original SHARED seed-83, 50-technician, concurrency-five cold run returned HTTP 409 at final audit. Detailed client timing and reason were not retained. Later audit evidence does not erase the failure. Frozen artifacts ran concurrently on one Windows workstation, limiting timing generalization. Browser coverage is INSERTION only and does not establish production concurrency performance.

Reproduce: install reportlab, matplotlib and pymupdf for Python 3. Run `py -3 infra/build-scheduler-algorithm-comparison.py` to regenerate both formats and shared PNG figures. Run it with `--check` to reconcile raw evidence and verify report content, images and links. Run `py -3 infra/test_scheduler_algorithm_comparison.py` for targeted regression checks. Raw evidence remains immutable.

Retained summary

Experiment manifest

Validation records

Illustrative fixture

Full field validation

September 24 daily solver protocol